

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total) Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

This program will conduct spectroscopy on a supernova using the “B&C” Spectrograph on the Bok 90” telescope at Kitt Peak, Arizona. Since supernovae need to be observed within approximately thirty days of peak brightness, the exact scientific goal of the project will not be known until the observations are conducted in Mid-October 2026. However, we can anticipate that by analyzing the spectra using SNID-Sage software, we can observe several key scientific insights regardless of the specific target selected: If we observe the supernova early enough, the spectrum will allow the team to classify the supernova type by identifying the presence or absence of defining spectral features, such as Hydrogen and Helium lines. Additionally, by measuring the Doppler shift of the supernova target’s spectral lines, we can possibly estimate the radial velocity and expansion speed of the supernova ejecta. Finally, the spectroscopy will reveal the relative abundances of various elements forged inside the star or during the explosion. Although the primary goal will become more clear during Mid-October, all of these ideas are fascinating and unique and contribute to a deeper understanding of supernovae and supernovae spectra.

Telescope Requested (0.5 points): BOK 90”

Instrument (0.5 points): “B&C” Spectrograph

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

At the time of our observation (Oct. 17-19), the sun will have an RA of around 14 hours, while the moon's RA will be between 18 and 20. This is not ideal for our observations as this means the moon will be reaching its peak right around sunset, however via the Catalina Sky Survey we can anticipate the brightness on these dates, from 38% to 58% illuminated as the weekend progresses. Thus, it would be more ideal to conduct our observations on October 17th if possible, as this is when the moon will be least bright and will have the least interference with our observations. The moon will set between 11:00PM and 1:00AM on all of these dates, so it may be advantageous to observe a supernova with an RA around 22-24 or 0-3 hours as these will be crossing the meridian while both the sun and moon are below the horizon.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

[Recently Discovered SN](#)

[Current Magnitude Tracking via Rochester Astro](#)

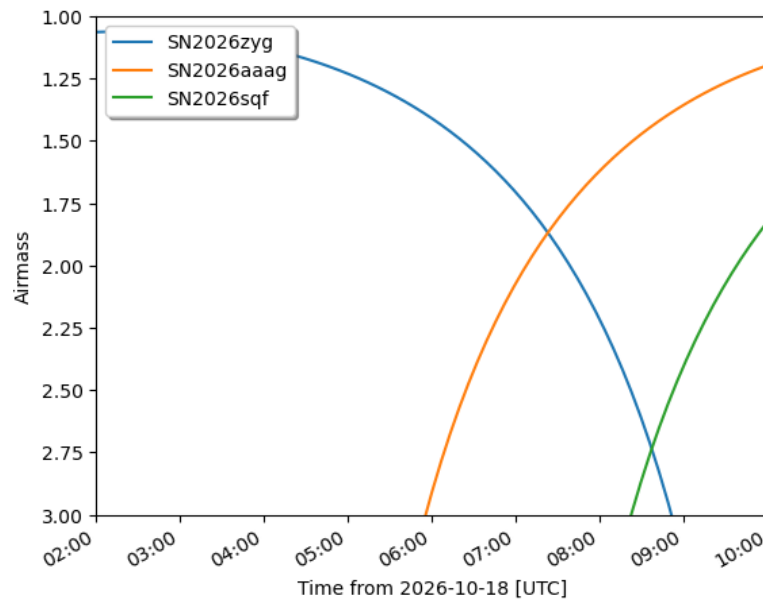
[UTC to Tucson Time](#)

MMT LOCATION: Latitude=+31d 41m 19.6s Longitude= 110d 53m 04.4s

Constellations highly visible in Oct: Cygnus, Pegasus, Cassiopeia, Cephus, Andromeda, Ursa Major/Minor

Constellations only visible at beginning/end of night: Aquarius, Capricornus, Cetus, Cygnus, Aquila, Lyra, Taurus, Auriga, Orion, Lynx

Object Name	Right Ascension	Declination	V magnitude at peak	Peak Date	Object Type	Visibility
SN 2026sqf	140.673888	53.37	12.9	7/28	SN Ia	ursa major
SN 2026aaag	102.05	51.23555556	18.08	8/29	SN type SN II	Lynx
<u>SN 2026zyg</u>	308.8761	51.5900	18.049	8/28	SN II	Cygnus



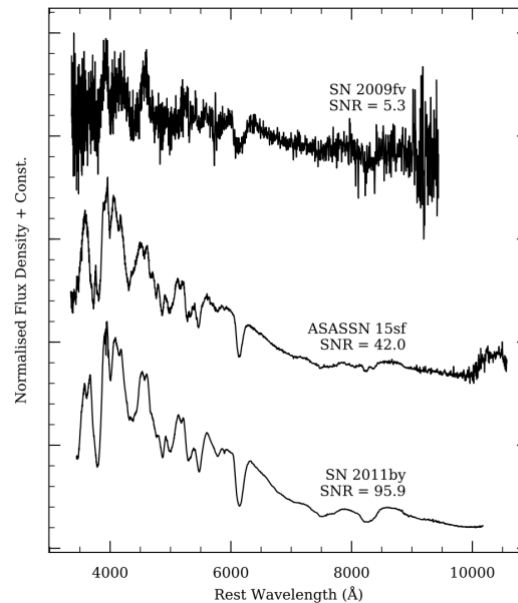
The plot shows the airmass for several of the targets outlined in the table for the night of October 17th from 7:00PM to 3:00AM Tucson time. The line in orange would be a better choice for a target in our case because it reaches peak later in the night when the moon will be set, and it gets very high in the sky which improves viewing with less airmass.

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is

necessary for your observations? (2 points)

In order to discern distinct spectral features like Hydrogen and Helium within



the supernova, we will want a minimum signal to noise ratio of five, ideally higher. This is based on a paper from Stahl et. al (2020) which observed Type Ia supernovae at varying signal to noise ratios and illustrated the visibility of spectral features for each. They used a 3 meter telescope, and in the paper they cite an SNR of 8.3 with an exposure time of 1200 seconds for one of their observations, and some much higher SNRs for the same exposure time. With a 2.3 meter telescope and probably a similarly bright source, we would likely need around 1200 seconds of exposure time to get an SNR of 5. Another reason that we chose a SNR of five is supported by a paper by [Fortino et. al](#) (2026), who calculate the minimum minimum spectral resolution and signal-to-noise ratio (SNR) required for the classification of SN subtypes. They state that supernovae subtypes can be discovered at as low of an SNR as five. For higher precision or fainter targets, we may need higher spectral resolutions and/or higher SNR, increasing exposure time.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

Taking a spectrum of the supernova will allow us to measure chemical absorption/emission lines, which are used in the categorization of the supernova based on the elements present from their features. For example, if there are no Hydrogen features in the spectrum, we will know that it is a Type I supernova, and the ionized Silicon (Si II) feature is an indicator of a type Ia supernova specifically. Using the SNID-Sage program will fit the spectrum and easily identify these features, while computations like optical depth can give us an idea of the abundances of certain elements within the supernova. We can estimate the radial velocity from the Doppler shift of certain known elemental features.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

If we are unable to gather our own spectroscopic data, we may be able to collect similar observations from projects that have been conducted by science teams at the University of Arizona recently. For example, Professor Nathan Smith does regular supernova spectroscopy using the same instrument, including data as recent as August of this year, so we could ask him to use some of his observations if we cannot get our own. That research group took spectra of a type Ic broad-line supernova earlier this year, which would be a unique and interesting science target for an alternative project.

Notes from Meeting with Dr. Milne

- October 17-19 observing only
- Maybe characterize SN if a new one goes off
- Most interesting SN science is within 30ish days of explosion (past peak)
- 1.5 airmass as general restriction
 - Don't go below -20-28 declination
- Sun will be at 14 RA in October, so we want to observe RA ~18+
- find other examples of what we want our supernova to be like
- Eventually run SNID-sage spectral fitter on our spectrum to identify type of supernova and fit
 - For 1A: measure velocity of silicon line
 - SNID will tell us interesting science information regardless of type

Possible Observing Days (on Bok):

- "You'll want your target to be more than 25° away from the moon at the very least, ie more than 1 to 2 hrs of RA and 15° to 25° of Dec away, and unless your target is very bright, perhaps a lot more than that"
- None of these days are on the full moon

- [Moon location](#), dates are in 0:00 UT so it will switch over to next day before we do at -5UT

October 2 - 3

Friday, Saturday

MOON: Illumination = 0.67-0.57, RA=05 06 40.98, DEC=+27 46 42.6

RA=06 10 56.40, DEC=+27 44 10.2

October 17 - 19

Saturday, Sunday, Monday

MOON: Illumination = 0.35-0.53, RA=18 24 30.79, DEC=-27 22 56.9

RA=19 16 50.06, DEC=-25 37 55.3

RA=20 07 29.03, DEC=-22 44 21.1